

Humphrey Curtis

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London, UK · Open to relocation

ABOUT

Research Engineer and Applied Scientist specialising in human-centred AI for accessibility and socially disadvantaged populations. I hold a PhD from King's College London and received the ACM SIGCHI Outstanding Dissertation Award. Previously, I was a Research Intern at Microsoft Research and have authored 15 peer-reviewed publications, including two ACM Best Paper Awards, an article in Nature Scientific Reports and a Routledge book chapter. My PhD focused on wearable and discreet AAC for people with post-stroke aphasia: through participatory design with disadvantaged communities across London, I investigated how smartwatches and smart badges can serve as communication safety nets, while also identifying where technologies such as smartglasses fall short because of cognitive load. I built tools across smartwatch, smart badge, iOS and HoloLens platforms, co-designing them with more than 200 stroke survivors. At Microsoft Research, I designed LLM-powered meeting agents for Microsoft Teams to help people with chronic illness stay connected in work and social life. I am now building Advokit, a GenAI-enhanced toolkit co-designed with aphasia communities to support access to public services, from benefit navigation to language assistance, using schema-driven LLM pipelines and adversarial guardrails; it is now deployed nationally by the UK Stroke Association and the DWP.

PROFESSIONAL EXPERIENCE

King's College London

London, UK

Research Associate, Department of Informatics, Human-Centred Computing Group

11/2025–Present

Supervised by Dr Filip Bircanin and Dr Timothy Neate

- Designed the LLM prompting layer and schema-driven agentic architecture for *Advokit* (advokit.uk) — a GenAI welfare-navigation toolkit where each page regenerates from a JSON schema co-editable by users and the LLM. Engineered structured reasoning chains, adversarial stress-tests and safety guardrails for a production system serving vulnerable users. Built a Next.js + MongoDB application with an LLM drafting assistant and safety monitoring.
- Built evaluation infrastructure (800 visitors, 2,849 page views) surfacing LLM failure modes, driving redesign toward text-box-free and agentic interactions. Translated R&D in stakeholder workshops with the Stroke Association and DWP; adopted for UK-wide rollout.

Microsoft Research

Seattle, USA

Ability Team; Research Intern

06/2024–09/2024

Mentored by Dr Martez Mott

- Designed and evaluated LLM-powered agentic meeting-assistant prototypes for Microsoft Teams, supporting chronically ill users pre-, during and post-call. First-authored CHI '25 paper; co-authored MSR tech report on AAC & agentic AI.
- Led the largest study of its kind (N=55) on video-calling experiences of chronically ill people; mixed-methods synthesis produced 8 design recommendations, presented to product and engineering teams.

King's College London, PhD Researcher

London, UK | 2021–2026

- Designed, built and deployed ML-driven communication tools and assistive technology across iOS, watchOS, e-ink (Badger2040) and mixed reality (HoloLens); co-designed with 200+ stroke survivors, clinicians and SLTs over four years (open-sourcing on GitHub).
- Developed evaluation frameworks for communicative confidence, felt understanding and social presence. Research contributed to a £2M UKRI Future Leaders Fellowship.

Lucky Beans Childcare Limited

London, UK

Freelance Web Development

06/2019–09/2019

Plenum Group

London, UK

Interim Executive Search

06/2018–09/2018

Santander

London, UK

Work Experience; Corporate Finance (Healthcare)

06/2016–09/2016

EuroABS

London, UK

Work Experience

06/2015–09/2015

EDUCATION

King's College London PhD in Computer Science, Department of Informatics (Awarded 2026) Supervised by Dr Timothy Neate and Dr Rita Borgo. Deployable wearable communication tools for people with aphasia: iOS apps, watchOS, e-ink (Badger2040) and mixed reality (HoloLens). Prototypes evaluated across 4 years of co-design with 200+ stroke survivors, SLTs and clinicians. Smartwatch and smartbadge emerged as the most viable — people valued personalisation, discretion and partner-awareness. Research contributed to a £2M UKRI Future Leaders Fellowship. Departmental profile: www.kcl.ac.uk/people/humphrey-curtis .	London, UK 09/2021–2026
King's College London MSc Advanced Computer Science; Distinction	London, UK 09/2020–09/2021
University of Bristol MSc Computer Science; Merit	Bristol, UK 09/2019–09/2020
Institut d'Études Politiques de Grenoble Year Abroad; Distinction	Grenoble, France 09/2017–09/2018
University of Exeter BA Politics, Philosophy and Economics; First Class	Exeter, UK 2015–2019
Dulwich College Secondary Education	London, UK 2009–2015

AWARDS & SCHOLARSHIPS

2026 ACM SIGCHI Outstanding Dissertation Award – Top doctoral thesis in human–computer interaction.
2026 King's Education Award Nomination, Colleagues' Choice – Nominated by colleagues in NMES faculty; award recognises meaningful contributions to education at King's College London.
2025 Ken Rothman Award – Health Research From Home.
2025 King's Doctoral College Fund – Awarded funds to travel and present at CHI 2025.
2024 Best Paper Nomination, ACM ASSETS – Premier assistive tech venue (top 5 of ~220 submissions).
2024 NMES PhD Research Competition Award – Faculty NMES PhD competition.
2023 Best Student Paper Award, ACM ASSETS – Premier assistive tech venue (top 2 of ~180 submissions).
2023 Best Paper Award, ACM CHI – Premier venue in HCI (top 1% of ~3,500 submissions).
2022 SIGACCESS Travel Scholarship – Funds to travel and present at ASSETS 2022.
2021 KCL PhD Studentship – Departmental scholarship to cover PhD, expenses and living costs.
2021 Academic Prize – Highest grade average across the entire MSc Advanced Computing cohort; top student by overall performance.

TECHNICAL SKILLS



AI & LLM: Prompt engineering (adversarial stress-testing, guardrail design, reasoning chains); schema-driven agentic pipelines; evaluation & monitoring; safety design for vulnerable users; on-device ML
Research Methods: Participatory design; mixed-methods; usability testing; diary studies; A/B testing; survey design; conversation analysis
Programming: Python; Swift; TypeScript/JavaScript (Next.js, React); HTML/CSS; C; SQL
Web & Infrastructure: REST APIs; MongoDB; Firebase; Vercel; Docker; Git; Linux
Design & Accessibility: Figma; WCAG; D3.js; DaVinci Resolve; Shotcut; Affinity Suite
Authoring: LaTeX; Microsoft Office; Emacs; Vim

PUBLICATIONS

ORCID | Google Scholar

Journal Articles

- J1** Curtis, H., & Neate, T. (2025). **Making smartglasses accessible: perspectives and prototypes from co-design with people with aphasia.** *Scientific Reports*, 15, 38309. DOI: 10.1038/s41598-025-22253-2

Technology Reports

- TR1** Beneteau, E., *et al.* (2026). **Adults Using AAC to Inform the Design of Mimetic Agentic AI.** *Microsoft Research Tech Report*. Link. (Second author; contributed participatory design insights to agentic AI systems).

Conference Papers

- C1** Curtis, H., Jenkins, A., Gentry, A., Zacharek, S., McVicker, S., Bircanin, F., & Neate, T. (2026). **“Computer Says No”: Disabled Welfare Experiences and Envisioned Futures Under AI Governance.** Proceedings of the ACM CHI Conference on Human Factors in Computing Systems (CHI ’26), Barcelona, Spain. DOI: 10.1145/3772318.3791025
- C2** Jenkins, A., Bagchi, P., Curtis, H., Seymour, W., Coté, M., & Such, J. (2026). **‘It literally feeds on data’: Co-designing Privacy Conscious and Trustworthy LLM Dialogues with End Users.** Proceedings of the ACM Conference on Designing Interactive Systems (DIS ’26), Singapore.
- C3** Curtis, H., Bircanin, F., & Neate, T. (2025). **“An Old Bastard in Bright Orange Satin!”: Zuzenna’s Aphasia Diary and Lessons Learned from DIY Augmentative and Alternative Communication.** Proceedings of the ACM SIGACCESS Conference on Computers and Accessibility (ASSETS ’25), Denver, USA. DOI: 10.1145/3663547.3746369
- C4** Mo, T., Curtis, H., & Neate, T. (2025). **Life as an International Computer Science PhD Student with Cerebral Palsy.** Proceedings of the ACM SIGACCESS Conference on Computers and Accessibility (ASSETS ’25), Denver, USA. DOI: 10.1145/3663547.3749835
- C5** Curtis, H., Beneteau, E., Cutrell, E., Ford, D., Junuzovic, S., Paradiso, A., Tang, J., & Mott, M. (2025). **“I use video calling in all areas of my life”: Understanding the Video Calling Experiences of Chronically Ill People.** Proceedings of the ACM CHI Conference on Human Factors in Computing Systems (CHI ’25), Yokohama, Japan. DOI: 10.1145/3706598.3713193
- C6**  Curtis, H., & Neate, T. (2024). **Looking Past Screens: Exploring Mixed Reality and Discreet AAC Devices.** Proceedings of the ACM SIGACCESS Conference on Computers and Accessibility (ASSETS ’24), St John’s, Canada. DOI: 10.1145/3663548.3675655
- C7** Curtis, H., & Neate, T. (2024). **Beyond Repairing with Electronic Speech: Towards Embodied Communication and Assistive Technology.** Proceedings of the ACM CHI Conference on Human Factors in Computing Systems (CHI ’24), Honolulu, USA. DOI: 10.1145/3613904.3642274
- C8** Curtis, H., Lau, Y., & Neate, T. (2024). **Breaking Badge: Augmenting Communication with Wearable AAC Smartbadges and Displays.** Proceedings of the ACM CHI Conference on Human Factors in Computing Systems (CHI ’24), Honolulu, USA. DOI: 10.1145/3613904.3642327
- C9**  Curtis, H., & Neate, T. (2023). **Watch Your Language: Using Smartwatches to Support Communication.** Proceedings of the ACM SIGACCESS Conference on Computers and Accessibility (ASSETS ’23), New York, USA. DOI: 10.1145/3597638.3608379
- C10**  Curtis, H., You, Z., Deary, W., Tudoreanu, M., & Neate, T. (2023). **Envisioning the (In)Visibility of Discreet and Wearable AAC Devices.** Proceedings of the ACM CHI Conference on Human Factors in Computing Systems (CHI ’23), Hamburg, Germany. DOI: 10.1145/3544548.3580936
- C11** Curtis, H., Neate, T., & Vazquez Gonzalez, C. (2022). **State of the Art in AAC: A Systematic Review and Taxonomy.** Proceedings of the ACM SIGACCESS Conference on Computers and Accessibility (ASSETS ’22), Athens, Greece. DOI: 10.1145/3517428.3544810

Under Review

- U1** Curtis, H., Gentry, A., Zacharek, S., McVicker, S., Neate, T., & Bircanin, F. (2026). **Prompting Change in the Welfare State: Co-designing an LLM-Assisted Toolkit of Welfare Stories and Tips with Aphasia Communities.** Submitted to the ACM SIGACCESS Conference on Computers and Accessibility (ASSETS ’26), Porto, Portugal. *Under review.*

Book Chapters

- B1** Curtis, H. & Neate, T. (2026). **Future Directions of AAC: Co-design of Wearable Communication Technologies with Users with Aphasia**. In A. Caute & A. Roper (Eds.), *Augmentative and Alternative Communication in Aphasia*. Routledge (Taylor & Francis).

Late-Breaking Work

- L1** Sattar, S., Curtis, H., & Neate, T. (2024). **Public Assistive Displays: Employing Public Interactive Displays to Improve Public Transport Access 4All**. Late-Breaking Work, ACM CHI Conference on Human Factors in Computing Systems (CHI '24), Honolulu, USA. DOI: 10.1145/3613905.3650890

Workshop Papers

- W1** Neate, T., Vincenzi, B., Barros Pena, B., Nevsky, A., Lewthwaite, S., Desai, S., Loizides, F., Curtis, H., & Bircanin, F. (2025). **Accessibility in Austerity: Formulating Strategies for Accessibility Research in Constrained Times**. Proceedings of the 38th International BCS Human-Computer Interaction Conference (BCS HCI '25). DOI: 10.14236/ewic/BCSHCI2025.4

Doctoral Consortium

- D1** Curtis, H. (2022). **Wearable, Discreet, Augmentative and Alternative Communication**. Doctoral Consortium, BCS Human-Computer Interaction (BCS HCI '22), Keele University, UK. DOI: 10.14236/ewic/HCI2022.54

TEACHING EXPERIENCE

6CCS3HCI: Human-Computer Interaction, King's College London	London, UK
Teaching assistant for labs and marker under Dr Timothy Neate, Dr Georgia Panagiotidou & Dr Petr Slovak	2021–2026
7CCSMSDV: Simulation and Data Visualization, King's College London	London, UK
Teaching assistant for labs and marker under Dr Rita Borgo	2022–2025
King's Undergraduate Research Fellowship	London, UK
Research Mentor for Master's and Undergraduate summer thesis projects	05/2023–08/2023

SERVICE & VOLUNTEERING

ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '26)	Porto, Portugal
Accessibility Co-Chair; AI Advisory Committee Member	2026
Aphasia Re-Connect, Roberta Williams Speech and Language Therapy Centre	London, UK
Weekly Charity Volunteer (under Dr Sally McVicker)	2021–Present
British Computer Society (BCS) Human-Computer Interaction Conference	Cardiff, Wales
Associate Chair	2025
Cockney CHAI	London, UK
Web Chair; designed and maintained the conference website	2025
ACM SIGACCESS Conference on Computers and Accessibility (ASSETS)	Athens; New York; St John's
Student Volunteer (2022–2023); Student Volunteer Chair and Research Mentor (2024)	2022–2024
ACM CHI Conference on Human Factors in Computing Systems	New Orleans, USA
Student Volunteer	05/2022

REVIEWING EXPERIENCE

ACM CHI Conference on Human Factors in Computing Systems (CHI)	2023–Present
ACM Symposium on User Interface Software and Technology (UIST)	2023–Present
ACM SIGACCESS Conference on Computers and Accessibility (ASSETS)	2025–Present
ACM Transactions on Accessible Computing (TACCESS)	2025–Present

BCS Conference on Human-Computer Interaction (BCS HCI)	2025
Augmentative and Alternative Communication (AAC Journal)	2024
ACM Transactions on Graphics (TOG)	2023, 2024
European Journal of Special Needs Education	2023
ACM Nordic Conference on Human-Computer Interaction (NordiCHI)	2023, 2024
ACM CHI Conference on Play (CHI PLAY)	2023

INVITED TALKS & MEDIA

<i>Advokit: Co-Designing Inclusive AI Tools for People with Aphasia</i> Stroke Association , Invited Talk	May 2026
<i>Advokit: AI-Enhanced Digital Access to Welfare Services for Disabled People</i> Department for Work and Pensions (DWP) , Invited Talk	May 2026
<i>The Importance of Ethics in HCI, AI and Assistive Technology</i> University of Oklahoma , HCI module guest lecture	March 2025
<i>Wearable and Discreet AAC Devices to Support Communication</i> Health Research From Home Showcase , Google HQ, King's Cross	February 2025
<i>Surveying Video Calling Experiences and Envisioning Digital Twins to Support Chronically Ill People</i> Microsoft Research , Ability Team — End-of-Internship Talk	September 2024
<i>Participatory Design & AAC: Looking Beyond Prominent Tech for Speech</i> Aphasia Summit , City, University of London	September 2024
<i>Not Everyone Needs a Device Like Stephen Hawking: Looking Beyond Prominent Technologies for Speech</i> HCID Day , City, University of London	September 2024
<i>Interview on UK Disability History Month and Personal Research</i> Enabled in Academia	December 2023
<i>Improving the Accessibility of London Transport</i> Transport for London (TfL)	November 2023
<i>Interview on Wearable AAC Devices</i> Electronic Specifier Magazine	August 2023

INTERESTS

Reading and Film: I am interested in literature, philosophy, history and cinema. Find me at: Goodreads and Letterboxd.
Sport: I have yet to find a sport I do not enjoy and golf remains the primary one. Represented King's College London and the University of Exeter in BUCS golf. Represented Spencer CC in cricket (4th nationally, 2012) and coached at Battersea Ironsides CC.

Societies and Leadership: Member of the University of Bristol Computer Science Society (2019–20). Former member of PPE Society and Exeter Entrepreneurs Society; served as a Study Abroad Mentor. Completed Excode (2018) and participated in the International Student Initiative at Sciences Po Grenoble (2017–18).

Challenges and Awards: Duke of Edinburgh Gold, Silver and Bronze Awards. Completed National Citizen Service (The Challenge, 2014).